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Distributed Temperature Survey Guided Coiled Tubing Stimulation of the First CO₂ Injector Well in UAE

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Abstract

This paper outlines the successful application of distributed temperature sensing (DTS) and fiber-optic-equipped coiled tubing (FOE-CT) in stimulating the first Carbon Dioxide (CO₂) injector well in the UAE. A Real-Time Assisted Matrix Stimulation Engineering (RTA-MSE) workflow enabled accurate injection profiling over a 600-ft perforated interval, optimizing acid placement and volume. The treatment resulted in a more than five-fold increase in injectivity, with DTS confirming uniform post-treatment acid distribution. To mitigate coiled tubing (CT) exposure to the CO₂ environment and avoid integrity risks, the wellbore fluid was initially displaced back to the formation using fresh water. In a single CT run, five DTS passes were recorded: DTS-1 for geothermal baseline, DTS-2 during freshwater injection to define the pre-treatment profile, and DTS-3 warm back to confirm flow distribution. Based on this data, the 28% hydrochloric acid treatment, enhanced with a nitrified stage, was optimized and pumped. DTS-4 and DTS-5—post-treatment injection and warm back—validated stimulation effectiveness. The CT was then safely retrieved, concluding the combined intervention and evaluation operation. This CO₂ injector well—the first of its kind in the Middle East—was drilled into a saline aquifer. Post-drilling, it achieved an initial injectivity of 7 bpm at 1,000 psi surface injection pressure. Distributed Temperature Sensing (DTS) profiling identified three distinct injection zones across the 600-ft perforated interval. Zones 2 and 3, located in the upper section, showed limited injectivity, which correlated with petrophysical interpretation and water injection simulation. In contrast, Zone 1, the lowermost section, exhibited uniform injection. Given the vertical well architecture, a nitrified acid stage was incorporated into the treatment to enhance coverage of the upper zones. Post-stimulation DTS data confirmed a significantly improved and uniform injection profile across all zones. As a result, the well's injectivity index increased by a factor of 5. This case validates the effectiveness of fiber-optic-enabled coiled tubing with real-time DTS in optimizing acid placement and improving zonal coverage in challenging CO₂ injection environments.

The methodology offers a replicable stimulation workflow for similar injector wells across the region, advancing stimulation design and monitoring practices in the petroleum industry. This paper presents the first documented case in the UAE of using DTS and downhole measurements to evaluate and optimize acid stimulation in a vertical CO₂ injector. By profiling injection across a 600-ft perforated interval in real-

time, the study demonstrates how DTS-enabled workflows can ensure uniform stimulation and significantly enhance injectivity, offering a new benchmark for matrix acidizing in CO₂ injection wells across the industry.

Introduction

ADNOC (Abu Dhabi National Oil Company) is leading a groundbreaking initiative in carbon capture and storage (CCS) with the launch of the world's first fully sequestered CO₂ injection well in a carbonate saline aquifer in Abu Dhabi. This project, part of ADNOC's multi-billion-dollar decarbonization action plan, aims to inject and permanently store at least 18,000 tons of CO₂ annually, initially captured from local UAE operations. The injection site was selected using advanced 3D seismic surveys and subsurface modeling, ensuring optimal geological conditions for long-term CO₂ storage (Xu 2022, Xu 2023). This initiative builds on ADNOC's earlier success with the Al Reyadah facility, which has the capacity to capture up to 800,000 tons of CO₂ per year. ADNOC's broader CCS strategy includes capacity to safely sequester 10 million tonnes of CO₂ per year by 2030.

These efforts align with ADNOC's commitment to achieving net-zero emissions by 2045 and achieve near-zero methane emissions by 2030, positioning the company as a regional and global leader in sustainable energy transformation.

Reservoir Description

The saline aquifer chosen as a reservoir for CO₂ sequestration is a prominent carbonate unit of Paleocene to early Eocene age, extensively developed across the Arabian Peninsula, including the United Arab Emirates, Saudi Arabia, Oman, and Qatar (Temitope 2016, Parra 2017). It represents a key stratigraphic and hydrogeological unit, characterized by its widespread distribution, distinct lithological features, and significant role in regional aquifer systems. The formation predominantly consists of dolomitic limestone and dolomite, interbedded with chalky wackestones and packstones. These rocks are typically light tan to gray and contain abundant microfossils, especially larger benthic foraminifera, which are crucial for biostratigraphic correlation and paleoenvironmental reconstruction. The aquifer is particularly notable for its vuggy limestones and localized sandstone layers, both of which enhance its permeability and groundwater storage capacity. The uppermost dolomitic layers are often highly porous and may include thin anhydrite beds.

The saline aquifer is considered a promising candidate for CO₂ sequestration due to a combination of geological, structural, and petrophysical characteristics that make it suitable for long-term carbon storage:

1. **Porosity and Permeability:** The formation is composed of dolomitic limestone and dolomite, which have undergone karstification and fracturing. These processes have enhanced the secondary porosity (such as vugs and dissolution cavities), allowing it to store large volumes of CO₂. The interconnected pore spaces also provide good permeability, enabling efficient injection and migration of CO₂ within the reservoir.
2. **Thick and Laterally Extensive:** The saline aquifer is regionally extensive across the Arabian Peninsula and can reach thicknesses of several hundred meters, providing a large storage volume. This makes it ideal for industrial-scale CO₂ sequestration projects.
3. **Deep Burial Depth:** In many areas, the formation lies at sufficient depths (typically >800 meters), where CO₂ can be stored in a supercritical state. This state is denser and more stable, allowing for more efficient storage and reduced risk of leakage.
4. **Effective Cap Rock:** The overlying formation contains impermeable anhydrite and shale layers. These act as a seal or cap rock, preventing the upward migration of CO₂ and ensuring long-term containment.
5. **Geological Stability:** The region has low seismic activity and stable tectonics, reducing the risk of fault reactivation or cap rock breach, which are critical for safe CO₂ storage.

1st CO₂ Sequestration Well – Well Completion Details

The subject well is completed as a cased-perforated completion targeting the saline aquifer introduced in previous section. Well completion details are provided in Fig. 1 (Mason 2024). The target saline aquifer is around 900 ft vertical thickness of which 600 ft are perforated in the 9 5/8" casing section (4,450ft-5,050 ft). The petrophysical characteristics of this saline aquifer are provided in Fig 2. As can be seen, the saline aquifer indicates average porosity of around 30% and average permeability of 80-100 md. However, there is a 40-50ft section of low permeability interval within the 600 ft perforated interval as highlighted by the boxed area.

Prior to commencing CO₂ injection in this well it was decided to obtain baseline water injectivity information. To ensure that water injectivity is uniform across the perforated interval, it was decided to conduct a coiled tubing deployed stimulation job to remove any drilling damage and to enhance the injectivity of the low permeability interval identified in the petrophysical log. In the following sections, we describe in detail the planning, execution and data analysis of the stimulation job carried out using a coiled tubing string equipped with fiber optic cable.

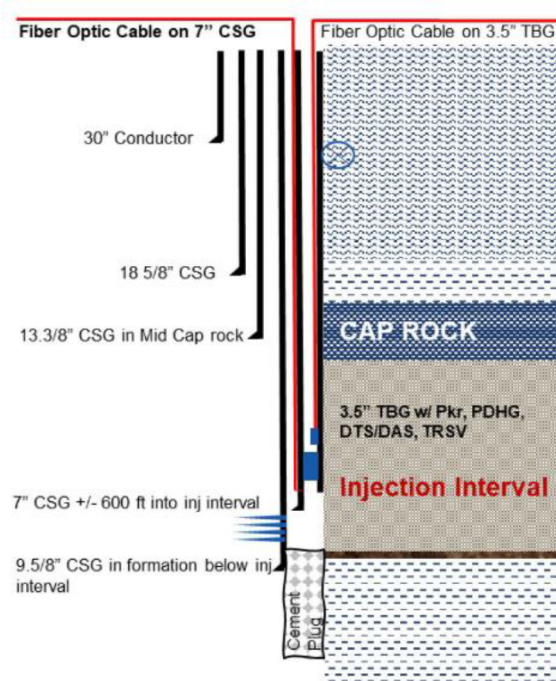


Figure 1—CO₂ Sequestration Well Completion Diagram

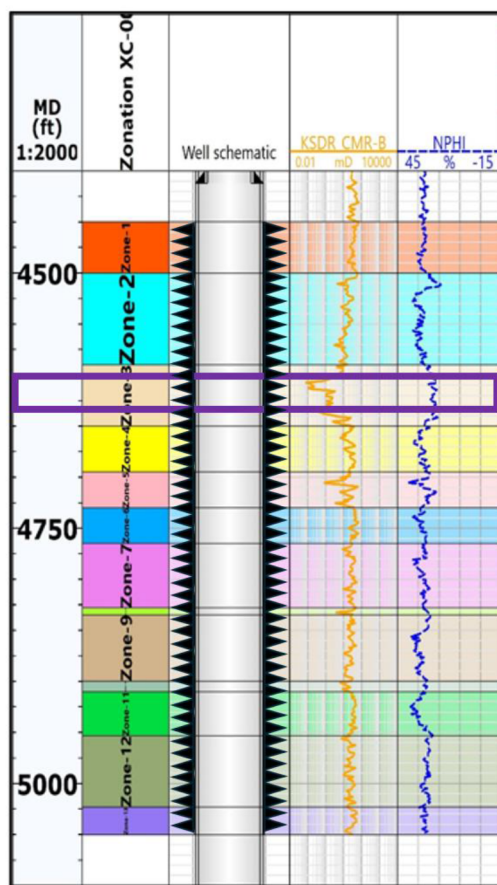


Figure 2—Petrophysical characteristics of target saline aquifer

Coiled Tubing Stimulation – Job Planning and Execution

The success of acidizing stimulation interventions depends on how well the damage and reservoir characteristics are understood and considered when designing and performing the treatment. With the advent of real-time monitoring capabilities such as the ones provided by the Fiber Optic Equipped Coiled Tubing (FOE-CT) system, it is now possible to mitigate any issue/deviation from the initial design while still in hole, to optimize the stimulation outcome. As such, acidizing interventions have the following well-established workflow:

1. **Initial Treatment Design:** this is usually carried out using a flow placement simulator, which can estimate the impact of various pumping schedules, different stimulation/diverter fluids, as well as provide insight into the expected temperature evolution because of the acid-rock reaction.
2. **Treatment Execution:** Carry out the pumping operations as per the designed pump schedule recommended from initial treatment design step.
3. **Treatment Evaluation:** after every main pumping stage, engineers check how the formation reacts to the treatment using the Distributed Temperature Survey.
4. **Decision:** Depending on the treatment evaluation result, it is decided whether to proceed with the job as it was initially designed, or whether a modified approach is necessary to address issues/deviations from initial design identified during the treatment evaluation.
5. **Re-Design:** if required, a modified pump schedule is developed and the execution pursued. Note that "new design" does not necessarily mean that a totally different approach is identified; it can simply be the adjustment of acid/diverter volumes and/or the fine-tuning of where they are to be pumped based

on the assessment done during the treatment evaluation. The loop is done as many times as necessary to reach an optimum stimulation outcome.

6. **Final Treatment Evaluation:** Once the job is completed, all the treatment data is gathered for a full treatment evaluation. This includes but not limited to running the flow placement simulator again with modified schedule(s) and comparing it to the initial design for matches. The insight gained from this exercise (as well as issues addressed during the job execution) can then be used as a basis for future treatment designs (this can apply for any future intervention or more specifically for a design pertaining to the same field/well/region).

The above steps can be visualized as per [Fig. 3](#) and represent the Real-Time Assisted Matrix Stimulation Engineering (RTA-MSE) workflow. This approach has been documented by [Maheshwari \(2024\)](#) when stimulating tight formations in the UAE.

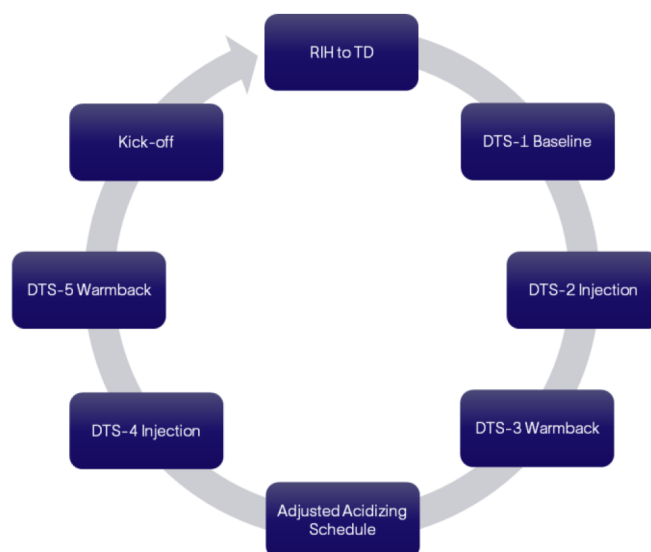


Figure 3—Real-Time Assisted Matrix Stimulation Engineering (RTA-MSE) workflow.

Acidizing interventions fully leverage the presence of a fiber-optics line within the coiled tubing pipe, as illustrated in [Fig 4](#), to help with the treatment evaluation. Advantages of FOE-CT are as follows:

- enable real-time downhole measurements (Pressure, Temperature, Casing Collar Locator, Gamma Ray, Tension and Compression sensors, as illustrated in [Fig 5](#)), which greatly helps with fluid placement, as described by [Franco \(2012\)](#)
- enables acquisition of Distributed Temperature Surveys (DTS), as shown by [Slama \(2024\)](#) and [Mason \(2024\)](#). These surveys provide the basis for the treatment evaluation and play a critical role toward the overall optimization of acidizing treatments and their ultimate outcome.

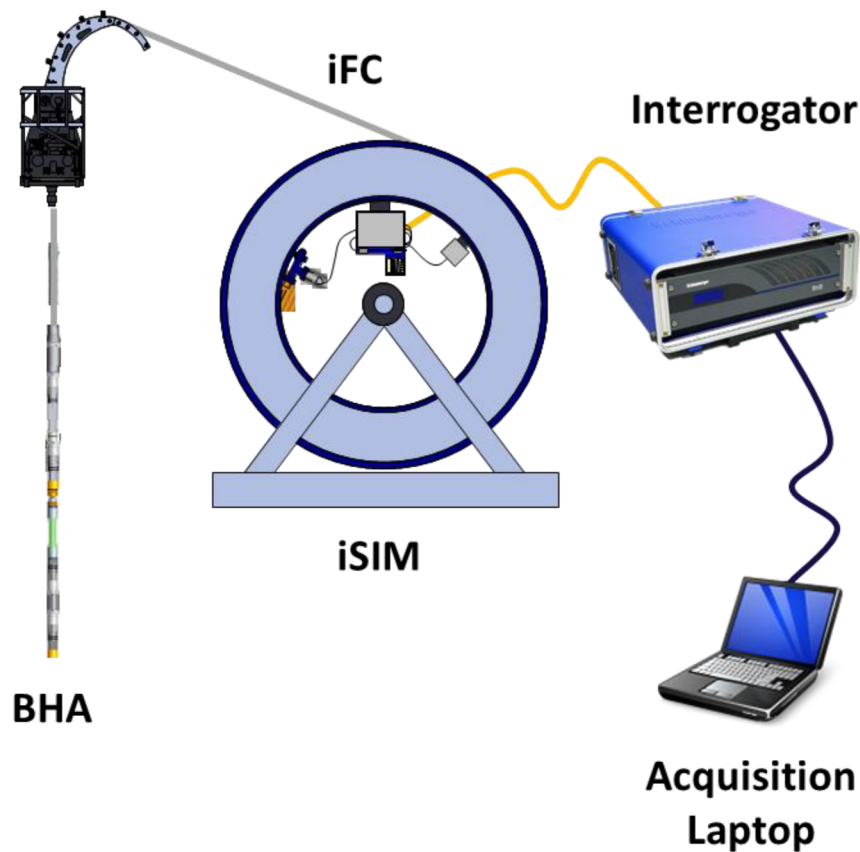


Figure 4—Fiber Optic Equipped Coiled Tubing

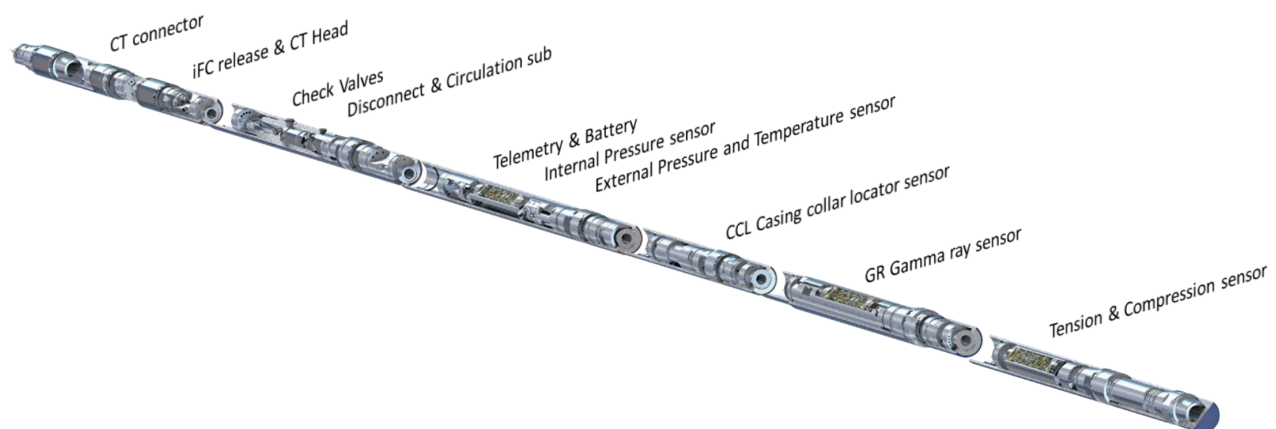


Figure 5—Downhole sensor package

Treatment Design

During the design phase and as part of the initial treatment design several simulations were run using as inputs the petrophysical properties of the formation such as the permeability and porosity profiles, the formation thickness and the well trajectory. The analysis concluded that using only a liquid phase would result in an overstimulation of the lowermost zones whereas using nitrified fluid would lead to a better stimulation of the upper zones, as illustrated in the below Fig 6.

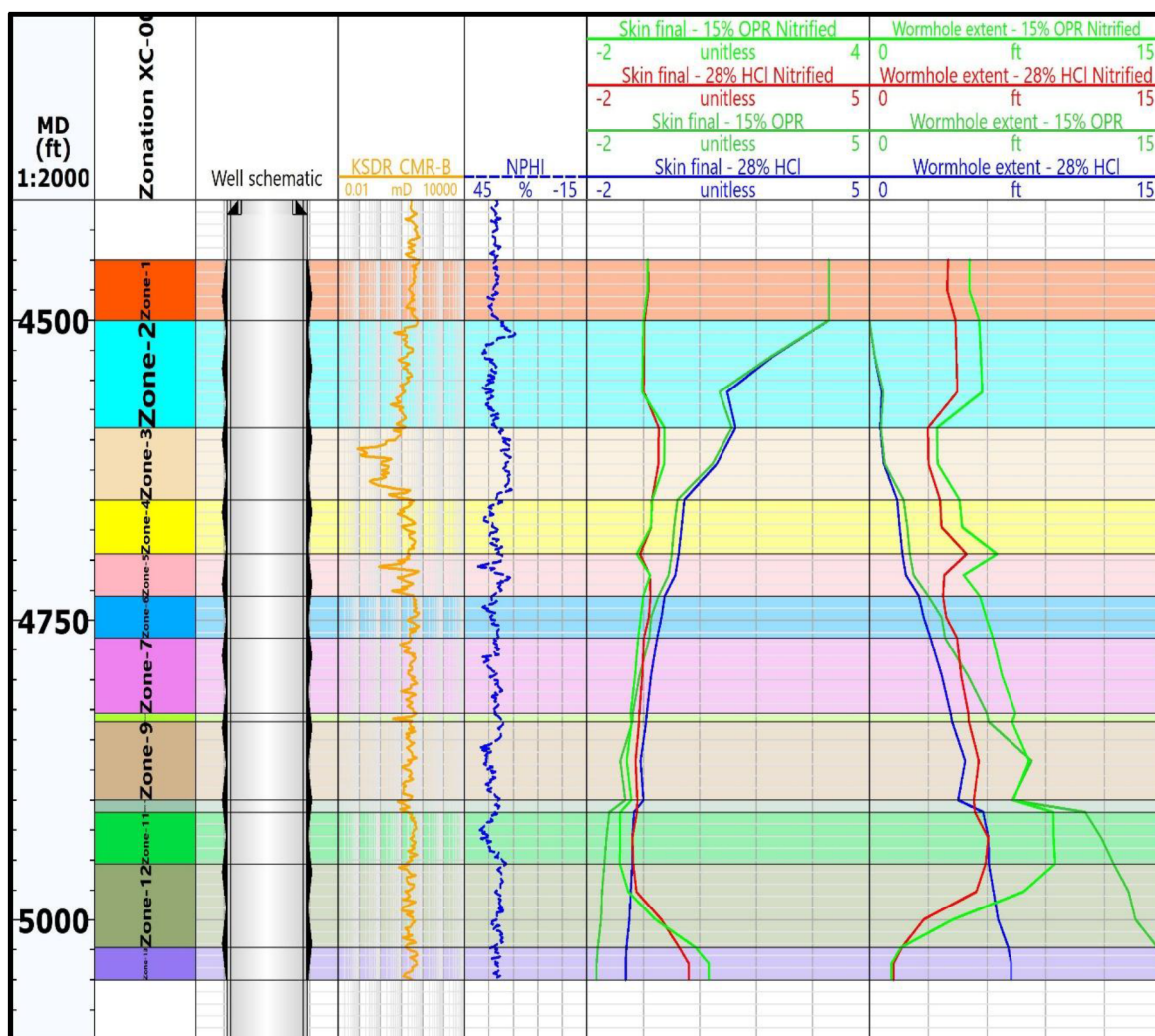


Figure 6—Matrix Stimulation Treatment Simulations

Execution phase

After completing equipment rig up and pressure tests, an injectivity test was carried out where a total of 134 barrels (bbl.) of fresh water was pumped at a maximum wellhead pressure of 742 psi using bullheading technique. Subsequently FOE-CT was run-in-hole (RIH) to 5175 ft (Run # 1). During RIH, depth correlation was carried out on three occasions. With CT stationed at 5175 ft baseline distributed temperature survey was performed (DTS-1). After baseline DTS-1 data acquisition was completed, pumping lines were configured to pump down via CT / Tubing annulus and a total of 1000 bbl. of fresh water at 6 barrels per minute (bpm) was pumped while acquiring DTS-2 Cooldown survey. DTS-3 warm back survey was acquired following the injection of water with FOE-CT stationed at 5175 ft. After acquiring DTS-3 data CT was pulled-out-of-the-hole (POOH), as illustrated in Fig 7.

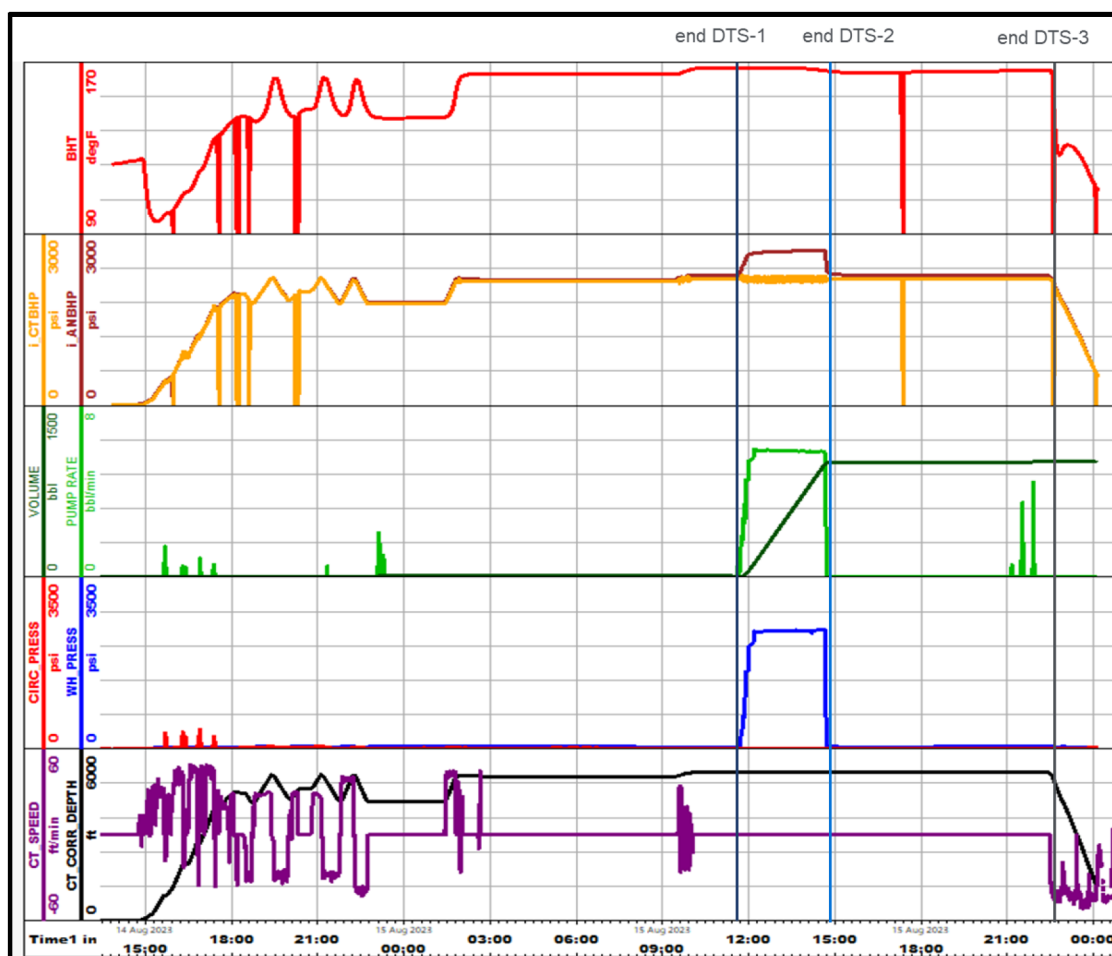


Figure 7—Surface and downhole parameters during Run # 1.

In Fig. 7, the upper track in red displays the temperature sensor readings. The second track from top (the brown and yellow lines) represent the downhole outside and inside pressures of the downhole sensor package. The light green and dark green lines on the third track represent the pumping rate and the total pumped volume. On the fourth track the red line and blue line represent the circulating pressure and the wellhead pressure. Finally, on the fifth track the black and purple lines represent the CT depth and the CT speed respectively.

After analyzing the pre-stimulation DTS data, some changes were made to the pump schedule as described later in this paper. FOE-CT was RIH to 5175 ft and depth correlation was confirmed twice (Run # 2). After this CT was POOH to 5,050 ft (bottom of the perf) at which point 100 bbl. of 10% Mutual Solvent (pre-flush) was pumped in the interval 5,050 – 4,450 ft while pumping at 1.8 bpm and POOH at 10 fpm. Once CT was at 4,450 ft nitrified acid (28% HCl at 1.2 bpm / Nitrogen at 750 standard cubic feet per minute) was pumped while CT was RIH from 4,450ft till 5,050 ft performing the main stimulation. Once the main stimulation was completed, FOE-CT was RIH to 5,175 ft and post-stimulation cooldown stage (DTS-4) was started by bull heading 850 bbl. of fresh water at 6.5 bpm through CT/tubing annulus for a period of 2.5 hrs. After completing the water injection, a post-stimulation warmback survey (DTS-5) was carried out. Once DTS-5 survey was completed FOE-CT was POOH, as illustrated in Fig 8.

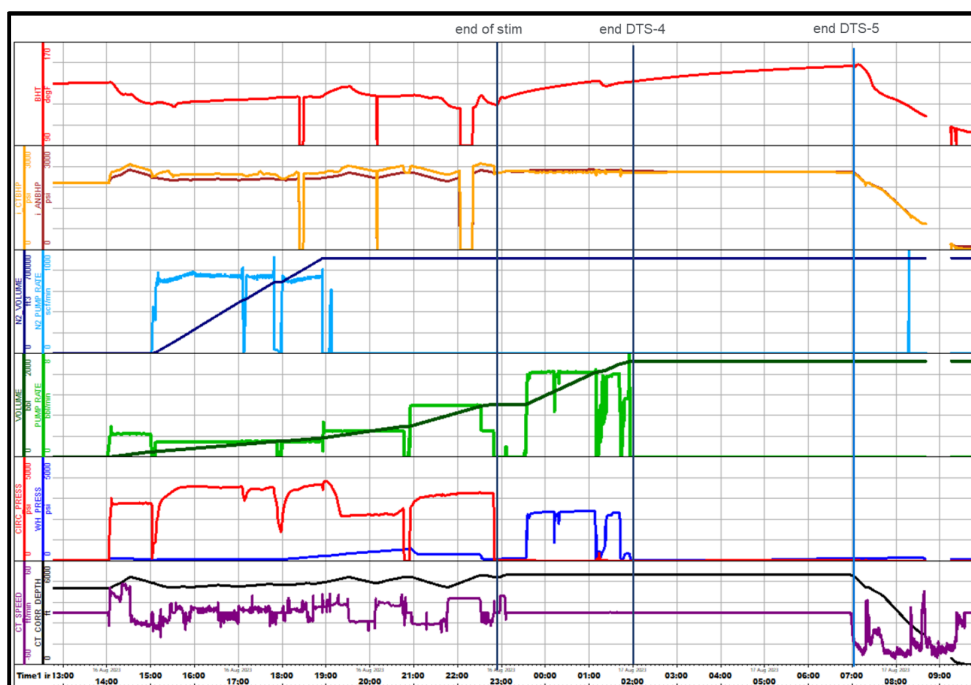


Figure 8—Surface and downhole parameters during run 2.

In Fig. 8 the upper track in red represents the temperature sensor readings. On the second track the brown and yellow lines represent the downhole outside and inside pressures of the downhole sensor package. The third track in light blue and dark blue lines represent nitrogen rate and volume. In the fourth track the light green and dark green represent the pumping rate and the total pumped volume. The fifth track with red line and blue line represent the circulating pressure and the wellhead pressure. Finally, the black and purple lines represent the CT depth and the CT speed respectively.

With CT at surface a long injection test was done by pumping 1,000 bbl. per day of fresh water at a pumping rate of 0.7 bpm during a period of 10 days.

Distributed Temperature Survey Analysis

Fig. 9 provides the compilation of various DTS surveys captured during the stimulation operation.

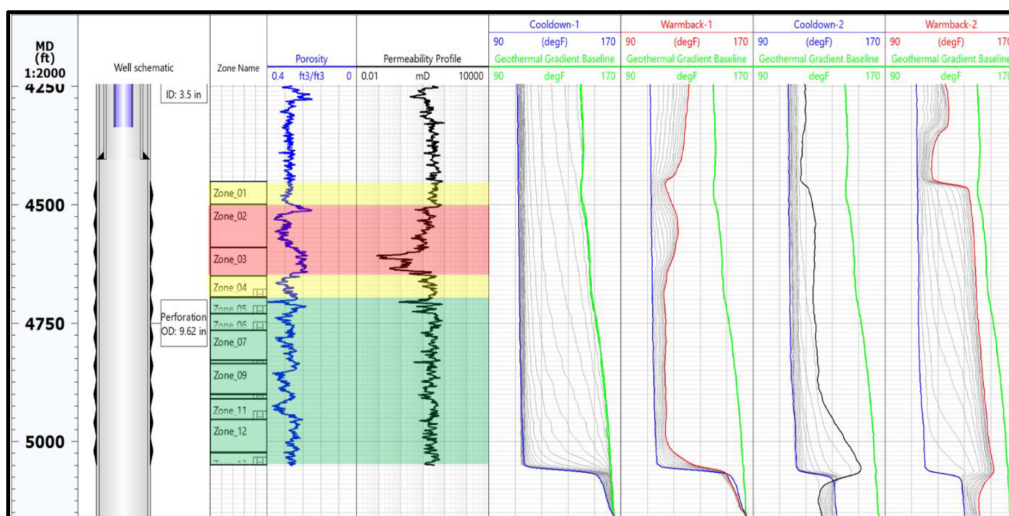


Figure 9—Distributed Temperature Survey During the job

From left to right, Fig. 9 displays the measured depth in the first track, followed by the well completion schematic at the interest zone. The next track displays the original thirteen zones that were reclassified by color. Green colored zones (zone 5-12) indicate the zones that are indicating the highest injectivity, the yellow zones (zone 1 and zone 4) indicate intermediate injectivity and the red zone (zones 2 and 3) as the lowest intake zones. The figure is complemented with the porosity and the permeability profile across the perforated interval. The last four tracks display DTS analysis. The first DTS track (DTS-1 and DTS-2) shows the geothermal (green line) and the temperature evolution during the injection (grey lines) and the last temperature profile achieved after the injection was concluded (blue line). The following track displays the warmback (DTS-3). One can notice that the temperature started to recover slowly (grey lines) until the last temperature profile (red line) indicating that the zones 2 & 3 (highlighted in the red area) showed faster temperature recovering. Thus, indicating that this zone presented lower water admission. This result matches the water injection simulation and petrophysical interpretation. Also, the lower most zones (5 to 13) present uniformity in the injection.

After this initial assessment, the acid treatment was redesigned by combining a nitrified stage to stimulate the upper zones (red and yellow ones) and a liquid stage to stimulate the green zone as described in the previous section.

After the stimulation treatment, the same process was followed, and the injection profile was acquired (DTS-4) followed by the warmback (DTS-5) profile. One can notice that the profiles are different from the ones before the stimulation. Most important, the warmback profile shows a uniform temperature recovery indicating a uniform injection across the 600ft perforated interval.

Results

Fig 10 shows the comparison of the injection rate parameters, before and after the main stimulation, and indicates an increase in injectivity index around 5.45-fold as estimated in Table 1. Data indicates that the injection rate increased from 5.87 bpm to 6.55 bpm while the downhole pressure was reduced from 2,716 psi to 2,308 psi, indicating elimination of the associated damage.

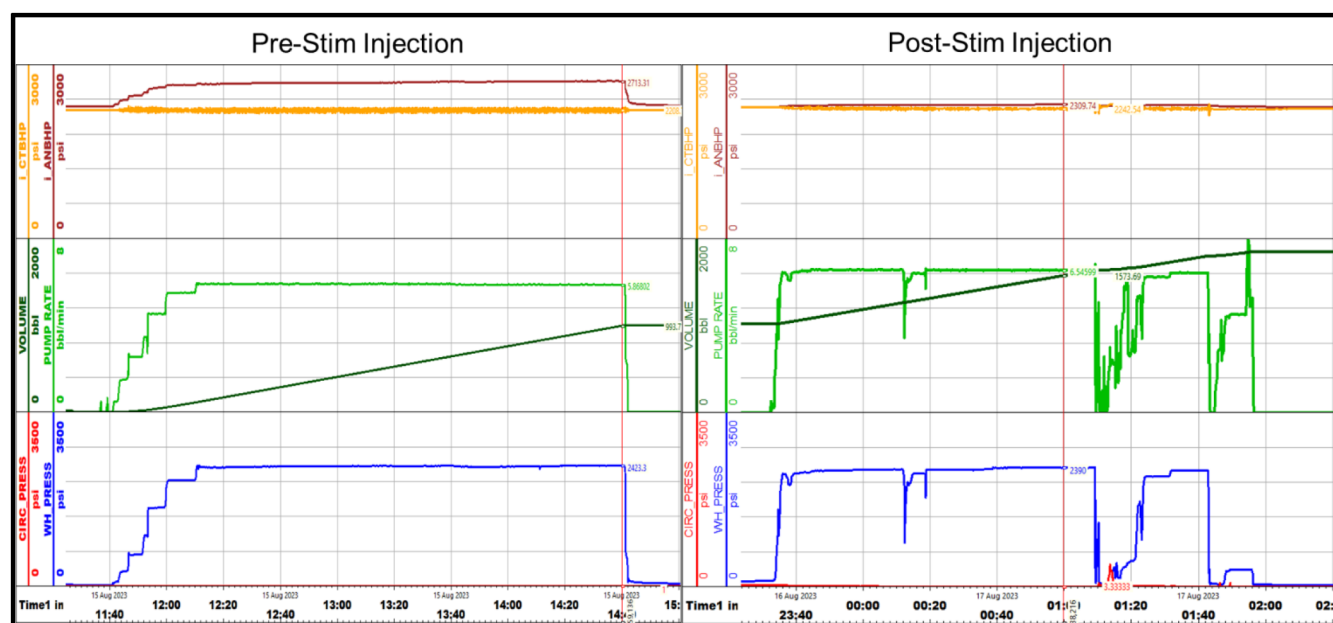


Figure 10—Comparison plots of the pre- and post-stimulation injection surface and downhole pressures vs pumping rates.

Table 1—Pre- and post-stimulation injectivity index calculations.

Pre-Stim			Post-Stim		
Rate	5.87	bpm	Rate	6.55	bpm
Daily Rate	8452.8	bpd	Daily Rate	9432	bpd
T. Vol.	1000	bbl	T. Vol.	800	bbl
Inj. Pres.	2716	psi	Inj. Pres.	2308	psi
BHP	2203	psi	BHP	2203	psi
Inj. Index	16.48	(bbl/d/psi)	Inj. Index	89.83	(bbl/d/psi)
Increase in Inj. Index		5.45 x			

Conclusions

Distributed temperature sensing (DTS) deployed on coiled tubing played a crucial role in providing real-time insights into fluid distribution under downhole conditions and helped characterize the reservoir by identifying tight zones, fluid intakes, and natural or induced fractures, while also verifying completion integrity. These insights enabled engineers to tailor stimulation strategies more effectively, plan future treatments with greater precision, maximize fluid penetration and diversion, and optimize both treatment volumes and fluid placement. More than 5-fold increase in injectivity was gained due to the precise placement of stimulation fluids aided by real-time downhole and DTS measurements. The stimulation job allowed the long-term water injection to be executed at substantially lower surface injection pressures in the saline aquifer to understand the reservoir behavior prior to commencing CO₂ injection.

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Nomenclature

BHA	Bottom hole assembly
BHP	Bottomhole pressure
CT	Coiled tubing
DTS	Distributed temperature survey
FOE-CT	Fiber optic equipped coiled tubing
Inj. Index	Injectivity Index (bbl/d/psi)
Inj. Pres.	Injection pressure (psi)
POOH	Pull out of hole
RIH	Run in hole
RTA-MSE	Real-Time Assisted Matrix Stimulation Engineering
T. Vol.	Total volume (bbl.)

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